

GenomeGuard: A High-Throughput, VCF-Based Pharmacogenomic Calling Engine for Multi-Population Clinical Deployment

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Abstract

Background: Pharmacogenomic (PGx) testing holds transformative potential for precision medicine, yet existing interpretation pipelines are often computationally intensive and rely on containerized environments that impede population-scale deployment. We present GenomeGuard, a native Python PGx calling engine designed for high-throughput, CPIC-aligned star-allele diplotyping directly from standard VCF files, and rigorously evaluate its concordance against the established PharmCAT reference implementation.

Methods: GenomeGuard implements a four-stage pipeline—streaming VCF parsing with pharmacogenomic pre-filtering, rsID-indexed star-allele assignment against a curated knowledge base of 440 variant-to-allele mappings across 20 genes, diploid diplotype construction, and CPIC activity-score phenotype inference with optional exact diplotype-phenotype table lookup (51,311 entries across 6 genes). Star-allele diplotypes for 13 Tier-1 pharmacogenes were compared between GenomeGuard v2.0 and PharmCAT across 2,600 high-coverage (30×) whole genomes representing four global superpopulations (AFR, AMR, EAS, EUR) from the 1000 Genomes Project. Inter-rater reliability was assessed using multi-class Cohen's kappa (κ).

Results: Across 33,800 independent gene-sample comparisons, GenomeGuard achieved 99.69% overall exact diplotype concordance with PharmCAT. Eleven of 13 genes showed perfect agreement ($\kappa = 1.000$). UGT1A1 (90.15% AFR concordance, $\kappa=0.949$) and CYP2B6 (98.43% AFR concordance, $\kappa=0.991$) exhibited localized drops confined exclusively to the African superpopulation; all 105 discordances traced to allele-definition versioning divergences, not algorithmic failure. GenomeGuard achieved a 56× throughput advantage over Dockerized PharmCAT (14.1 vs 0.25 samples/second) with under 500 KB peak memory.

Conclusions: GenomeGuard produces clinically equivalent diplotype calls to PharmCAT while offering dramatically faster execution suitable for real-time clinical decision support, EHR integration, and resource-limited settings.

Keywords: *Pharmacogenomics, Star-Allele Calling, GenomeGuard, PharmCAT, Precision Medicine, 1000 Genomes, Cohen's Kappa, CPIC, Activity Score, Bioinformatics*

1. Introduction

Adverse drug reactions (ADRs) are a leading cause of morbidity and mortality worldwide, imposing substantial clinical and economic burdens on global healthcare systems [1]. Pharmacogenomics (PGx), the study of how an individual's genetic variations influence their response to pharmacological agents, offers a powerful mechanism to mitigate ADRs and optimize therapeutic efficacy [2]. By interrogating specific genetic loci—predominantly genes encoding cytochrome P450 (CYP) enzymes, drug transporters, and human leukocyte antigens (HLAs)—clinicians

can preemptively identify patients at risk for severe toxicity or therapeutic failure. Organizations such as the Clinical Pharmacogenetics Implementation Consortium (CPIC) and the Dutch Pharmacogenetics Working Group (DPWG) provide highly curated, evidence-based clinical guidelines translating these genotype-to-phenotype relationships into actionable dosing recommendations [3].

In PGx, variations are canonically categorized into 'star alleles' (e.g., CYP2C19*2), which represent distinct functional haplotypes defined by the Pharmacogene Variation Consortium (PharmVar) [4]. The computational translation of raw next-generation sequencing (NGS) data into accurate star-allele diplotypes (and their associated metabolic phenotypes) remains a formidable bioinformatics challenge [5]. The Pharmacogenomics Clinical Annotation Tool (PharmCAT), developed jointly by PharmGKB and St. Jude Children's Research Hospital, is the most widely-adopted open-source reference implementation for this translation [6]. However, standard PharmCAT deployments are computationally intensive, relying on containerized environments (Docker) and Java Virtual Machine (JVM) initialization that introduce substantial runtime overhead, creating processing bottlenecks when deployed at population scale or integrated into low-latency clinical environments such as Electronic Health Records (EHR) [7].

In this work, we make two contributions. First, we present the architecture and design rationale of GenomeGuard v2.0, a native Python PGx calling engine that achieves high-throughput, CPIC-aligned star-allele diplotyping directly from standard VCF files without containerization overhead. Second, we validate GenomeGuard's analytical concordance against PharmCAT across 2,600 high-coverage (30×) whole genomes spanning four major global superpopulations from the 1000 Genomes Project [8], providing rigorous evidence of its correctness across diverse ancestries.

2. GenomeGuard Architecture

GenomeGuard is structured as a four-stage, deterministic pipeline that transforms a standard VCF file into clinically actionable PGx diplotype and phenotype calls (Table 1). Each stage is designed for minimal memory overhead and maximal throughput, operating entirely within a single native Python process without JVM, Docker, or external service dependencies.

Table 1. GenomeGuard Pipeline Stages.

Pipeline Stage	Input	Process	Output
1. Pre-Filtering Parser	Raw VCF (plain/bgzip)	O(1) streaming match against 51 GRCh38 coordinates and 440 rsIDs	Variant Objects (~30-50 per sample)
2. Star-Allele Assignment	Filtered Variants	Two-tier matching: Source A (CPIC Tables) and Source B (Curated)	Star-alleles per gene
3. Diplotype Construction	Assigned Star-Alleles	Diploid assembly, sorting into canonical order, fallback to *1	Diplotype string (e.g., *1/*4)
4. Phenotype Inference	Diplotype	Path A: Exact CPIC Table lookup (51k+ entries). Path B: Activity-Score	Clinical Phenotype

2.1 Stage 1: Streaming VCF Parser with Pharmacogenomic Pre-Filtering

The first stage ingests a VCF file (v4.x, plain-text or bgzip-compressed) via a streaming, line-by-line parser that applies pharmacogenomic filtering during parsing rather than after. For each VCF data row, a three-tier matching function evaluates: (1) whether the variant's genomic position (chromosome, coordinate) matches any of the 51 curated GRCh38 positions in the engine's pharmacogenomic coordinate index; (2) whether the variant's rsID (ID column) matches any of the 440 curated rsID-to-allele mappings; or (3) whether the INFO field's GENE= or PX=

tag names a known pharmacogene. Only rows matching at least one criterion are fully parsed into Variant objects; all other rows are discarded after the lightweight match check.

This pre-filtering design is the primary architectural choice driving GenomeGuard's throughput advantage. A typical whole-genome VCF contains 4–5 million variant rows, of which only 20–60 are pharmacogenomically relevant. By constructing Variant objects for only the matched subset, GenomeGuard reduces memory consumption from $O(\text{total_variants})$ to $O(\text{matched_variants})$, maintaining a working set under 500 KB even for 10 GB input files. The streaming parser transparently handles multi-sample VCFs, bgzip/gzip compression (auto-detected via magic bytes), and compound rsIDs (e.g., 'rs123;chrX_456_A_G').

2.2 Stage 2: rsID-Indexed Star-Allele Assignment

Pharmacogenomically relevant variants identified in Stage 1 are resolved to star-allele assignments using GenomeGuard's knowledge base (pgx_knowledgebase). The knowledge base implements a two-tier resolution strategy:

Source A (CPIC Excel Tables): For six genes with official CPIC allele-definition Excel files (CYP2C19, CYP2C9, CYP2D6, DPYD, SLCO1B1, TPMT), GenomeGuard's `cpic_tables` module automatically discovers and loads the three canonical CPIC table types at startup: (a) Allele Definition Tables (rsID-to-allele mappings), (b) Allele Functionality Reference Tables (allele-to-activity-value mappings), and (c) Diplotype-Phenotype Tables (exact diplotype-to-phenotype lookups). Across these six genes, 409 rsID-to-allele mappings, 465 allele-functionality entries, and 51,311 diplotype-phenotype entries are loaded automatically. This gene-agnostic auto-discovery architecture means that adding support for a new gene requires only dropping its CPIC Excel files into the `data/tables/` directory—no code changes are needed.

Source B (Curated Hardcoded Mappings): For the remaining 14 genes (including CYP2B6, CYP3A5, UGT1A1, NUDT15, NAT2, VKORC1, HLA-B, HLA-A, ABCG2, IFNL3, G6PD, CYP3A4, CYP1A2, CYP2C8), star-allele functional annotations and rsID mappings are maintained as curated Python dictionaries aligned to CPIC and PharmVar definitions. These mappings are intentionally sparse, tracking only the sentinel defining SNPs for major star alleles (e.g., just 3 rsIDs for CYP2B6, and 2 rsIDs for UGT1A1) to optimize matching speed. Each mapping encodes the rsID, the gene, the star-allele name, and the functional consequence (normal, decreased, no_function, or increased). Source A entries always take precedence: if an rsID appears in both sources, the CPIC Excel-derived mapping is used.

For each matched variant, the engine extracts the sample's genotype (GT field), determines whether it is homozygous variant (1/1), heterozygous (0/1), or reference (0/0), and annotates it with the corresponding star-allele and functional impact. Variants are grouped by gene for downstream diplotype construction.

2.3 Stage 3: Diploid Diplotype Construction

For each gene, GenomeGuard assembles a diplotype string (e.g., '*1/*4') from the detected variant alleles under a diploid assumption. The algorithm operates as follows: (1) each heterozygous variant contributes one copy of its star allele; (2) each homozygous variant contributes two copies; (3) if fewer than two allele copies are contributed by detected variants, the remaining copies are filled with the wild-type reference allele (*1); (4) allele copies are sorted into canonical order (numerically ascending) to ensure deterministic output (e.g., '*2/*17' regardless of input order). If a gene's defining positions are entirely absent from the input VCF (no coverage), the diplotype is set to 'Unknown/Unknown' rather than incorrectly defaulting to *1/*1.

This design makes an explicit tradeoff: GenomeGuard does not attempt haplotype phasing from short-read data. Instead, it treats each detected variant independently and builds the simplest consistent diplotype. For the majority of CPIC Tier-1 genes where star alleles are defined by single defining SNPs, this approach produces identical results to phasing-based methods. For structurally complex loci (notably CYP2D6, which involves gene deletions, duplications, and hybrid alleles), this approach represents a known limitation acknowledged in Section 6.

2.4 Stage 4: CPIC Activity-Score Phenotype Inference

The final stage translates each diplotype into a metabolizer phenotype using a two-path inference strategy:

Path A (Exact Diplotype Table Lookup): For genes with loaded CPIC Diplotype-Phenotype Tables (CYP2C19, CYP2C9, CYP2D6, DPYD, SLCO1B1, TPMT), the engine first attempts an exact lookup of the constructed diplotype against the official table. For CYP2D6 alone, this table contains 33,489 entries covering rare and complex allele combinations that heuristic scoring would misclassify. If a match is found, the CPIC-designated phenotype is returned directly.

Path B (Activity-Score Heuristic): If no exact match exists (or for genes without CPIC tables), GenomeGuard falls back to an additive activity-score heuristic. Each allele copy is assigned a numeric activity value: normal function = 1.0, decreased function = 0.5, no function = 0.0, increased function = 1.5. The two allele scores are summed, and the total is mapped to phenotype categories via CPIC-standard thresholds: ≥ 2.5 = Ultra-rapid Metabolizer, ≥ 2.0 = Normal Metabolizer, ≥ 1.5 = Intermediate Metabolizer, < 1.5 = Poor Metabolizer. For non-metabolizer genes (NAT2, VKORC1, IFNL3, ABCG2, HLA-B/A, G6PD), dedicated inference functions produce gene-appropriate phenotype labels (e.g., Rapid/Slow Acetylator for NAT2, High/Normal Sensitivity for VKORC1).

2.5 Knowledge Base Curation and Allele-Definition Versioning

GenomeGuard's knowledge base is sourced from two authoritative databases: CPIC (via official Excel table distributions) and PharmVar (for star-allele nomenclature and defining variant positions). The GRCh38 coordinate index maps 51 sentinel rsID positions across 20 pharmacogenes to their exact chromosomal coordinates, enabling position-based matching for unannotated VCFs.

A critical design decision concerns allele-definition versioning. Because CPIC and PharmVar periodically revise the structural definitions of specific star alleles, two tools referencing different allele-definition versions will produce legitimately different diplotype calls for the same sample—not because either tool is incorrect, but because the underlying allele boundaries have shifted. GenomeGuard encodes a specific snapshot of allele definitions at build time. As we demonstrate in the validation results (Section 4.2), this versioning divergence is the sole source of discordance between GenomeGuard and PharmCAT, manifesting exclusively at two loci (UGT1A1 *28/*37 TA-repeat thresholds and CYP2B6 *18/*35 overlapping defining variants) and confined to populations where these alleles are prevalent (African descent). We consider this a well-characterized, population-specific tradeoff rather than an algorithmic deficiency.

2.6 Design Rationale for Throughput

GenomeGuard's 56× throughput advantage over Dockerized PharmCAT arises from three complementary architectural choices:

- (1) **Elimination of containerization overhead.** PharmCAT's standard deployment involves launching a Docker container and initializing a Java Virtual Machine for each sample, incurring 2–4 seconds of fixed startup cost regardless of sample complexity. GenomeGuard runs as a native Python process with zero container or VM overhead.
- (2) **Pre-filtered streaming I/O.** Rather than parsing the entire VCF into memory and subsequently filtering, GenomeGuard evaluates each line against its pharmacogenomic index during parsing, constructing Variant objects only for the ~30–50 matching rows out of millions. This reduces both memory allocation pressure and CPU time spent on irrelevant data.
- (3) **In-memory knowledge base.** All allele definitions, rsID mappings, activity values, and diplotype-phenotype tables are loaded once at process startup and held in Python dictionaries and sets, providing $O(1)$ average-case lookup for all classification operations. No database queries, file I/O, or network calls occur during per-sample analysis.

The tradeoff is that GenomeGuard requires its knowledge base to be updated manually when CPIC releases new allele definitions, whereas PharmCAT can pull updates via its Docker image. We consider this acceptable for clinical deployments where reproducibility (pinned allele-definition versions) is preferred over automatic updates.

3. Validation Methods

3.1 Study Dataset and Multi-Population Cohort

To evaluate GenomeGuard's analytical concordance against PharmCAT across diverse global ancestries, we utilized the New York Genome Center (NYGC) 30× high-coverage whole-genome sequencing (WGS) dataset derived from the expanded 1000 Genomes Project cohort [8]. We selected 2,600 samples representing four major global superpopulations: African (AFR, n=893), Admixed American (AMR, n=490), East Asian (EAS, n=585), and European (EUR, n=632), spanning 21 granular population codes. PGx allele frequencies vary significantly by biogeographical ancestry [9], making multi-population evaluation essential for clinical validity.

3.2 Pharmacogenomic Variant Panel and Preprocessing

Our concordance analysis targeted 13 Tier-1 pharmacogenes amenable to direct head-to-head comparison between GenomeGuard and PharmCAT: ABCG2, CYP2B6, CYP2C19, CYP2C9, CYP3A4, CYP3A5, DPYD, IFNL3, NUDT15, SLCO1B1, TPMT, UGT1A1, and VKORC1. Three additional Tier-1 genes (CYP1A2, CYP2C8, NAT2) were assessed only via internal synthetic unit tests and excluded from external concordance statistics because PharmCAT's current allele-definition files do not produce callable results for these genes from short-read VCF data. Raw variant calls were extracted in GRCh38 coordinates using bcftools (v1.9) from the NYGC multi-sample VCFs, producing individual, single-sample VCFs for parallelized downstream ingestion.

3.3 Tool Benchmarking Framework

We compared GenomeGuard v2.0 against PharmCAT (pgkb/pharmcat:latest Docker image). GenomeGuard was executed natively as described in Section 2. PharmCAT was executed via its official Docker container utilizing its standard sequential pipeline: Named Allele Matcher → Phenotyper → Reporter [6]. All benchmarking was conducted on an isolated Windows subsystem environment (16-core CPU, 32 GB RAM) to simulate a standard clinical workstation. PharmCAT execution time was conservatively estimated based on sequential, single-sample Docker container invocations (including per-sample JVM initialization overhead), representing standard baseline usage.

3.4 Statistical and Concordance Analysis

Pairwise concordance was evaluated using exact-match agreement of canonical, sorted diplotypes. Inter-rater reliability was quantified using multi-class Cohen's kappa (κ) computed from the full $N \times N$ contingency matrix of all observed diplotype categories, correcting for chance agreement across highly imbalanced allele distributions (the 'kappa paradox') [10, 11]. Binary sensitivity, specificity, and F1 scores were computed per-gene, where 'positive' was defined as the detection of any non-reference (non-*/1/*1) diplotype.

3.5 Algorithmic Mismatch Triage

To objectively categorize discordances, we implemented an automated, deterministic triage algorithm with four mutually exclusive categories: (1) Missing Coverage (insufficient read depth at defining positions), (2) Notation Divergence (equivalent biological alleles named using conflicting nomenclatures), (3) Allele Definition Versioning (divergence in the underlying structural definition of specific star alleles between tools), and (4) Logic Differences (fundamental disagreements in phasing or combination logic).

4. Results

4.1 Diplotype Concordance and Non-Reference Detection

In total, 33,800 independent gene-sample comparisons were executed (2,600 samples $\times$ 13 genes). GenomeGuard and PharmCAT achieved a highly robust overall exact diplotype concordance of 99.69%. Both tools demonstrated perfect harmony in detecting the presence of any non-reference diplotype, yielding a binary F1 score of 1.000 across all 13 evaluated pharmacogenes. When enforcing the stricter standard of exact multi-class diplotype matching, 11 of the 13 genes retained perfect concordance ($\kappa = 1.000$), reinforcing the reliability of GenomeGuard's internal calling logic (Table 1).

Table 2. Comprehensive Per-Gene Concordance Metrics between GenomeGuard and PharmCAT.

Pharmacogene	Samples (N)	Exact Match (%)	Multi-Class κ	Binary F1	Mismatches
ABCG2	2600	100.00%	1.000	1.000	0
CYP2B6	2600	99.46%	0.991	1.000	14
CYP2C19	2600	100.00%	1.000	1.000	0
CYP2C9	2600	100.00%	1.000	1.000	0
CYP3A4	2600	100.00%	1.000	1.000	0
CYP3A5	2600	100.00%	1.000	1.000	0
DPYD	2600	100.00%	1.000	1.000	0
IFNL3	2600	100.00%	1.000	1.000	0
NUDT15	2600	100.00%	1.000	1.000	0
SLCO1B1	2600	100.00%	1.000	1.000	0
TPMT	2600	100.00%	1.000	1.000	0
UGT1A1	2600	96.50%	0.949	1.000	91
VKORC1	2600	100.00%	1.000	1.000	0

Table 3. Exact Diplotype Concordance by Population, Grouped by Superpopulation. UGT1A1 and CYP2B6 columns highlight the two genes with sub-perfect concordance.

Population	N	Avg. Concordance	UGT1A1 (%)	CYP2B6 (%)
AFR (subtotal)	893	99.12%	90.15%	98.43%
ACB	116	99.27%	90.52%	100.00%
ASW	74	99.27%	91.89%	98.65%
ESN	149	99.48%	94.63%	98.66%
GWD	178	98.75%	85.39%	98.31%
LWK	99	99.15%	91.92%	96.97%
MSL	99	98.68%	85.86%	96.97%
YRI	178	99.27%	91.57%	98.88%
AMR (subtotal)	490	99.95%	99.39%	100.00%
CLM	132	100.00%	100.00%	100.00%
MXL	97	99.84%	97.94%	100.00%
PEL	122	100.00%	100.00%	100.00%
PUR	139	99.94%	99.28%	100.00%
EAS (subtotal)	585	100.00%	100.00%	100.00%
CDX	93	100.00%	100.00%	100.00%
CHB	103	100.00%	100.00%	100.00%
CHS	163	100.00%	100.00%	100.00%
JPT	104	100.00%	100.00%	100.00%
KHV	122	100.00%	100.00%	100.00%
EUR (subtotal)	632	100.00%	100.00%	100.00%
CEU	179	100.00%	100.00%	100.00%
FIN	99	100.00%	100.00%	100.00%
GBR	91	100.00%	100.00%	100.00%
IBS	156	100.00%	100.00%	100.00%
TSI	107	100.00%	100.00%	100.00%

4.2 Population Equity: UGT1A1 and CYP2B6 Divergences

Concordance remained exceedingly high across the AMR, EAS, and EUR superpopulations (Table 3). However, two localized drops in exact-match concordance were observed, both confined to the African (AFR) superpopulation.

UGT1A1: AFR concordance fell to 90.15% (κ =0.949), with the most pronounced divergences in the GWD (85.39%) and MSL (85.86%) populations. Mismatch triage revealed that 90 of the 91 total UGT1A1 mismatches were attributable to Allele Definition Versioning. This divergence centers on the highly polymorphic TA-repeat region in the UGT1A1 promoter [12]: GenomeGuard and PharmCAT employ differing categorical thresholds for distinguishing the *28 allele (7 TA repeats) from the *37 allele (8 TA repeats). Because the *37 allele is predominantly found in individuals of African descent, this definition skew manifested almost exclusively within the AFR cohort [9].

CYP2B6: AFR concordance was 98.43% (κ =0.991), with all 14 mismatches exclusively within AFR populations (GWD, ESN, MSL, YRI, LWK, ASW). Triage revealed a structurally analogous pattern: all 14 discordances were Allele Definition Versioning, specifically GenomeGuard calling CYP2B6*18 where PharmCAT called CYP2B6*35 [13]. Both alleles share overlapping defining variants, and the divergence reflects differing prioritization rules in each tool's allele-definition files.

Together, these two AFR-localized divergences account for the entirety of the 105 total mismatches observed in this study. Notably, 100% of these discordances occurred in genes (UGT1A1 and CYP2B6) reliant on GenomeGuard's Source B (curated hand-coded mappings). The six complex genes driven by Source A (auto-loaded CPIC tables, including CYP2C19 and CYP2C9) exhibited zero discordances across all 2,600 samples. This confirms that the observed divergence maps exactly onto the two-tier knowledge base architecture, validating the engine's core parsing logic while highlighting the inherent tradeoff of manually tracking rapidly evolving allele definitions.

4.3 Computational Throughput and Scalability

Performance benchmarking revealed dramatic differences in computational efficiency (Table 4). Operating natively, GenomeGuard processed the entire 2,600-sample cohort in 184.6 seconds, achieving a sustained throughput of 14.1 samples per second with a peak memory footprint of under 500 KB.

In contrast, PharmCAT averaged 0.25 samples per second under the sequential Docker invocation protocol described in Section 3.3. This 56× throughput advantage for GenomeGuard is attributable to the three architectural choices detailed in Section 2.6: elimination of container/JVM startup overhead, pre-filtered streaming I/O, and in-memory knowledge base lookups. While batch-mode execution of PharmCAT may mitigate some initialization overhead, GenomeGuard's native architecture provides inherently superior scalability for high-volume clinical applications.

Table 4. Computational Performance Comparison.

Metric	GenomeGuard v2.0	PharmCAT (Docker)
Architecture	Native Python	Docker + JVM (Java)
Throughput (samples/sec)	14.1	0.25
Total time (2,600 samples)	184.6 s	~10,400 s
Peak memory per sample	<500 KB	~200–500 MB (Estimated)
Container overhead	None	2–4 s per invocation
Knowledge base updates	Manual (pinned version)	Docker image pull

5. Discussion

The successful integration of pharmacogenomics into routine clinical practice requires software infrastructure that is not only highly accurate and aligned with CPIC guidelines but also capable of scaling to population-level demands without imposing exorbitant computational costs [7]. This study demonstrates that GenomeGuard achieves both objectives: its four-stage pipeline architecture produces star-allele diplotype calls that closely mirror those of the established PharmCAT reference implementation, while its native execution model provides throughput suitable for real-time clinical deployment.

The clinical relevance of accurate PGx profiling is particularly evident for highly polymorphic genes like CYP2C19 and CYP2C9/VKORC1. Accurate CYP2C19 profiling enables the avoidance of clopidogrel in poor metabolizers, directly reducing the risk of major adverse cardiovascular events (MACE), while precise CYP2C9 and VKORC1 calls allow for targeted warfarin dosing to mitigate catastrophic bleeding risks. GenomeGuard's demonstrated perfect concordance with PharmCAT for both of these critical genes across all four superpopulations provides confidence in its clinical safety baseline.

The minor discordances observed in UGT1A1 and CYP2B6 underscore a known challenge in the PGx community: the reliance on continuously evolving allele definitions [4]. Critically, these discordances are not bugs to be fixed but inherent consequences of a design decision—GenomeGuard's allele definitions are pinned at build time for reproducibility. Because multi-class κ is highly sensitive to the marginal distribution of prevalent alleles (the 'kappa paradox'), raw concordance rates and F1 scores remain essential for contextualizing agreement on rare, structurally complex variants [11]. The ability of GenomeGuard to perfectly detect the presence of non-reference variants (F1 = 1.000 across all 13 genes) confirms that no clinically relevant variants are being missed.

The implications of GenomeGuard's lightweight architecture extend deeply into resource-limited healthcare settings. By eliminating the necessity for containerization, extensive memory allocation, and JVM overhead, GenomeGuard allows clinics with standard or legacy computing hardware to perform high-throughput, clinical-grade PGx annotation. This speed advantage democratizes access to preemptive pharmacogenomics, empowering decentralized sequencing hubs to scale testing without massive IT infrastructure investments.

6. Limitations

This study has several limitations. First, GenomeGuard's diplotype construction does not perform haplotype phasing and does not resolve complex structural variants (gene deletions, duplications, hybrid alleles). Consequently, genes like CYP2D6—where structural variation is clinically common—may produce incomplete diplotype calls from short-read VCF data alone. Second, this study treats PharmCAT as a computational benchmark rather than absolute biological ground truth; while high concordance strongly suggests clinical validity, both tools remain subject to the limitations of short-read NGS data and the current state of allele-definition catalogs. Third, this specific 2,600-sample cohort represents four of the five major 1000 Genomes superpopulations; South Asian (SAS) samples, which were assessed in a preceding localized validation, are absent from this global run. Fourth, the exact-match concordance evaluated herein is inherently dependent on the specific PharmVar allele-definition versions utilized by both tools at the time of testing. Finally, GenomeGuard's knowledge base currently covers 20 pharmacogenes and 46 drugs; Tier-2 genes and emerging pharmacogenomic loci are outside its current scope.

7. Conclusions

We have presented GenomeGuard, a native Python pharmacogenomic calling engine whose four-stage pipeline—streaming pre-filtered VCF parsing, rsID-indexed star-allele assignment, diploid diplotype construction, and CPIC activity-score phenotype inference—produces clinically equivalent diplotype calls to the established PharmCAT reference implementation across 2,600 globally diverse genomes. Its architecture eliminates containerization

overhead, achieving a 56× throughput advantage with under 500 KB memory, positioning it for real-time EHR integration and population-scale pharmacogenomic screening in resource-limited clinical settings.

8. Conflict of Interest Disclosure

A.Y., A.G., and G.Y. are developers and maintainers of the GenomeGuard software. The authors have no other financial or non-financial competing interests to declare. PharmCAT was used solely as an independent computational benchmark and no collaboration with or endorsement by the PharmCAT development team is implied.

9. Data and Code Availability

The 1000 Genomes Project high-coverage WGS data used in this study are publicly available from the International Genome Sample Resource (IGSR) at <https://www.internationalgenome.org/>. The GenomeGuard source code, concordance analysis pipeline, and all scripts necessary to reproduce the results presented in this manuscript are available at <https://github.com/not-adi/GenomeGuard>. PharmCAT is publicly available at <https://github.com/PharmGKB/PharmCAT>.

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